

DATA WITH BARAA

SQL COMPONENTS

Baraa Khatib Salkini
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SQL Course | SELECT Query



```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```

DATA WITH BARAA

SQL STATEMENT

```sql
-- Retrieve Customers Data

SELECT

    name,

    LOWER(country)

FROM customers

WHERE country = 'Italy'
```

Comment

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```

Clauses

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```

Keywords

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```

Function

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)
FROM customers
WHERE country = 'Italy'
```



```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

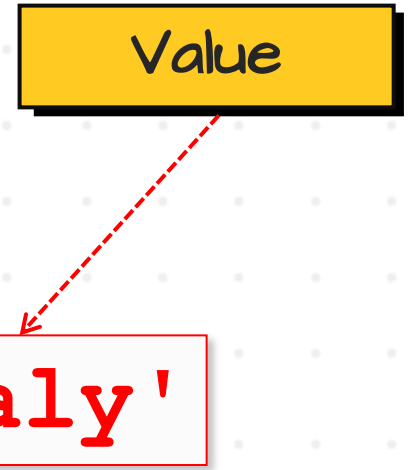
WHERE country = 'Italy'
```

Operator

```sql
-- Retrieve Customers Data
SELECT
    name,
    LOWER(country)

FROM customers

WHERE country = 'Italy'
```





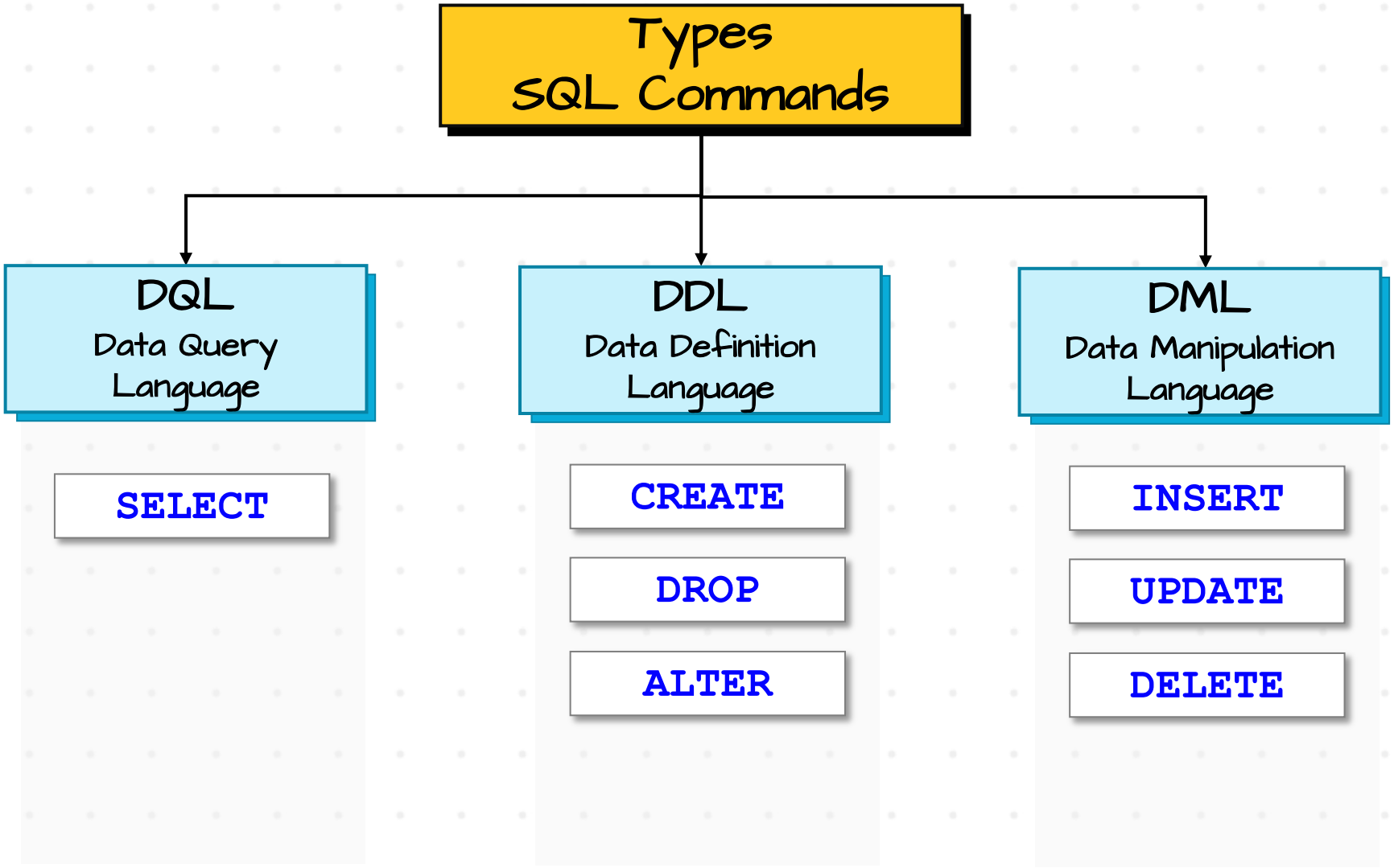
DATA WITH BARAA

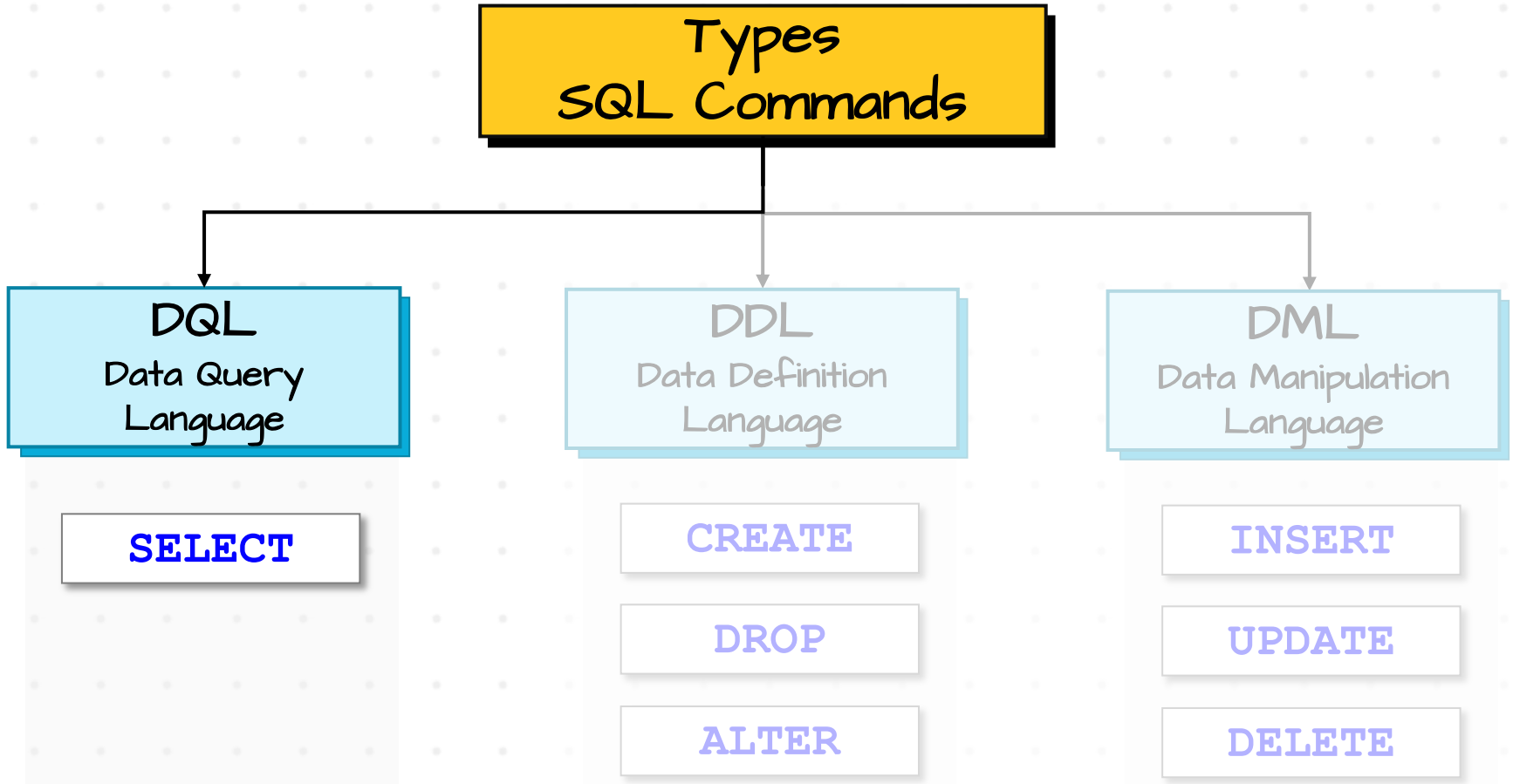
QUERY DATA

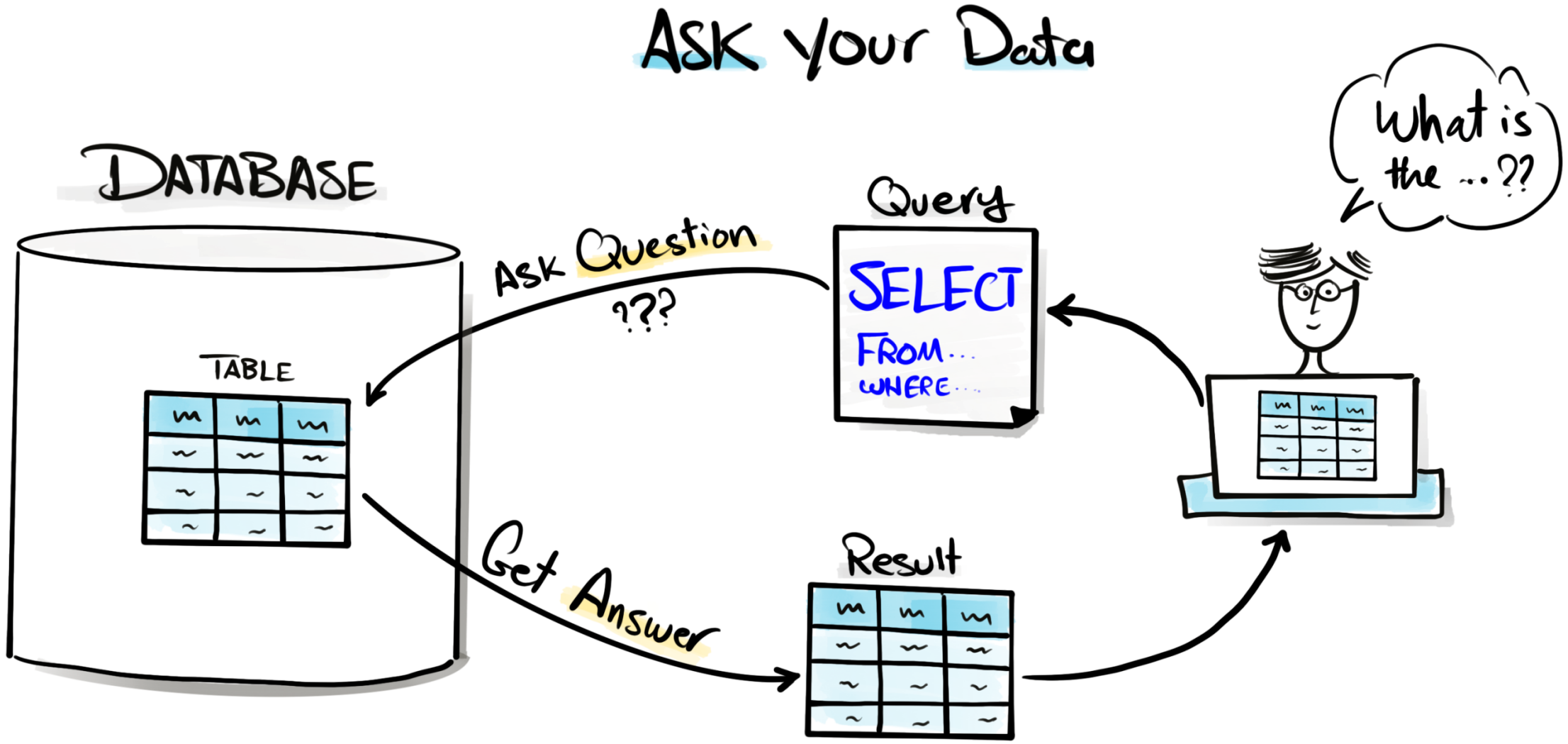
SELECT STATEMENT

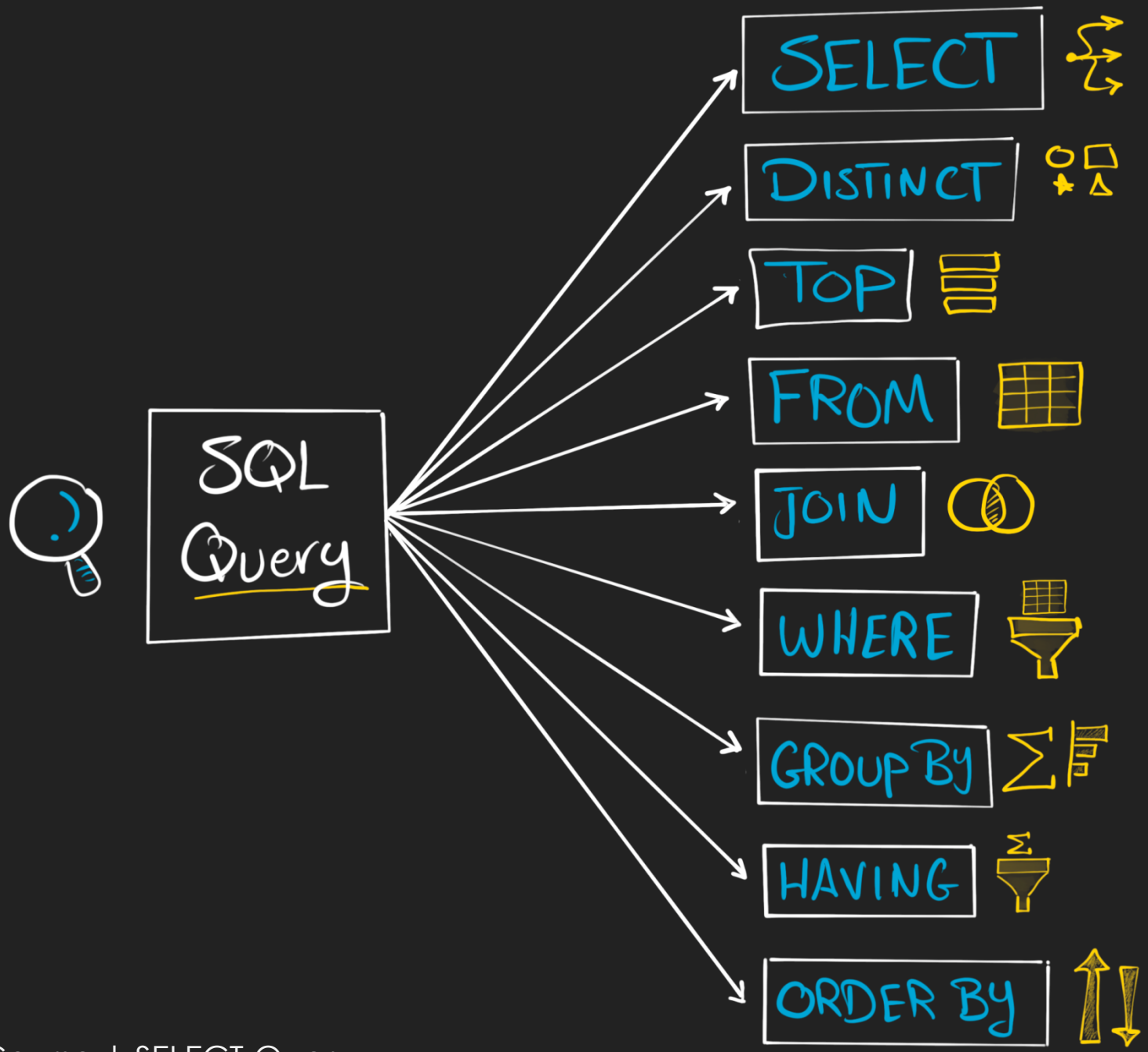
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Types
SQL Commands
DQL
Data Query Language
SELECT
DDL
Data Definition Language
CREATE
DROP
ALTER
DML
Data Manipulation Language
INSERT
UPDATE
DELETE


Types
SQL Commands
DQL
Data Query Language
SELECT
DDL
Data Definition Language
CREATE
DROP
ALTER
DML
Data Manipulation Language
INSERT
UPDATE
DELETE


ASK your Data
DATABASE
TABLE
Query
SELECT
FROM...
WHERE...
ASK Question
???
What is the ...??
Get Answer
Result



DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

DATA WITH BARAA

SELECT *

FROM Table

Select * (All)

Retrieves All Columns (Everything)

From

Tells SQL Where to find Your Data

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Keep All Columns!!

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

2 SELECT *

1 FROM Table

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Select Few Columns

Pick only the Columns You Need

instead of All.

```
SELECT
Col 1,
Col 2
FROM Table
```


DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

② SELECT
Col 1,
Col 2

① FROM Table

Keeps only Needed Columns

name	Country
Maria	Germany
John	USA
Georg	UK
Martin	Germany
Peter	USA

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

DATA WITH BARAA

```
SELECT *
FROM Table
WHERE Condition
```

Where



Filters Your Data based on a Condition

Score Higher than 500

Data with Baraa

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

3 SELECT *

1 FROM Table

2 WHERE Condition

Score > 500



id	name	Country	Score	
1	Maria	Germany	350	✗
2	John	USA	900	✓
3	Georg	UK	750	✓
4	Martin	Germany	500	✗
5	Peter	USA	0	✗

Data with Baraa

Database

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

```
SELECT *
FROM Table
ORDER BY Score DESC
```

Order By

Sort Your Data

- ASC
  - Lowest
  - Highest
- DESC
  - Highest
  - Lowest

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

3 SELECT *

1 FROM Table

2 ORDER BY Score DESC

Highest



Lowest

id	name	Country	Score
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
1	Maria	Germany	350
5	Peter	USA	0

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Lowest

Highest

id	name	Country	Score
4	Martin	Germany	500
1	Maria	Germany	350
3	Georg	UK	750
2	John	USA	900
5	Peter	USA	0

Highest

Lowest

Highest

Lowest

③ SELECT *

① FROM Table

② ORDER BY

Country ASC,

Score DESC

DATA WITH BARAA

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Group By

Combines rows with the Same Value

Aggregates a Column By another Column

Total Score By Country

```
SELECT
    Country,        -- Category
    SUM(score)      -- Aggregation
FROM Table
GROUP BY Country
```

Data with Baraa

Database



	id	name	Country	Score
1	1	Maria	Germany	350
2	2	John	USA	900
3	3	Georg	UK	750
4	4	Martin	Germany	500
5	5	Peter	USA	0

Σ

1	Germany	850
2	USA	900
3	UK	750

```
3 SELECT
    Country,
    SUM(score)
1 FROM Table
2 GROUP BY Country
```

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Having

Filters Data After Aggregation

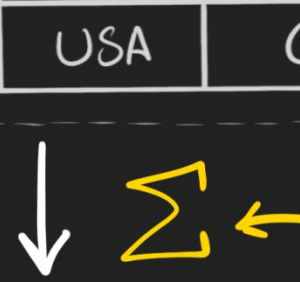
Can be used only with Group By

```
SELECT
    Country,
    SUM(score)
FROM Table
GROUP BY Country
HAVING SUM(score) > 800
```

DATA WITH BARAA

Database

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0



Germany	850
USA	900



Total Score > 800

4 SELECT
Country,
SUM(score)

1 FROM Table

2 GROUP BY Country

3 HAVING SUM(score) > 800

DATA WITH BARAA

Data with Baraa

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0



Filter Your Data

① Before Aggregation

② After Aggregation

```
SELECT
    Country,
    SUM(score)
FROM Table
WHERE Score > 400
GROUP BY Country
HAVING SUM(score) > 800
```

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

```
SELECT DISTINCT
    Col
FROM Table
```

Distinct

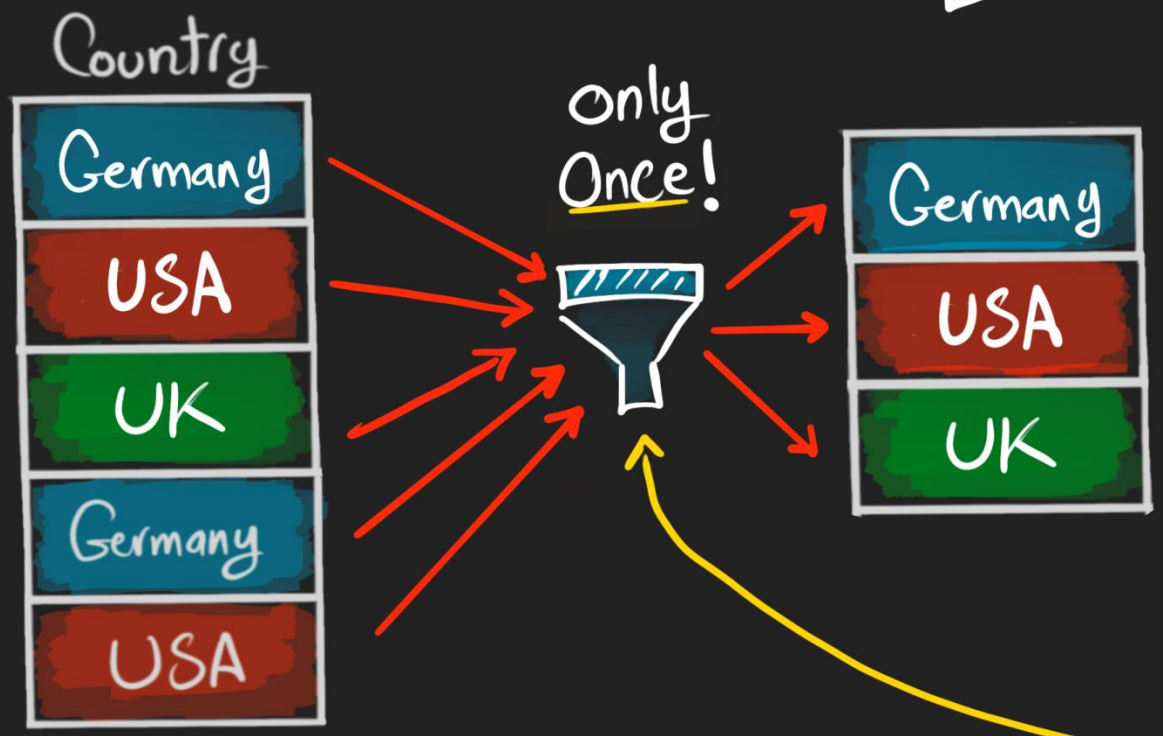
Removes Duplicates (Repeated Values)

Each Value appears only Once

DATA WITH BARAA

Database

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0



3

② SELECT DISTINCT Country

① FROM Table

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

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SELECT Top 3 *

FROM Table

Top (Limit)



Restrict the Number of Rows Returned

DATA WITH BARAA

Database



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

③ ② SELECT Top 3

\+

① FROM Table

	id	name	Country	Score
Row 1 →	1	Maria	Germany	350
Row 2 →	2	John	USA	900
Row 3 →	3	Georg	UK	750

Execute Order



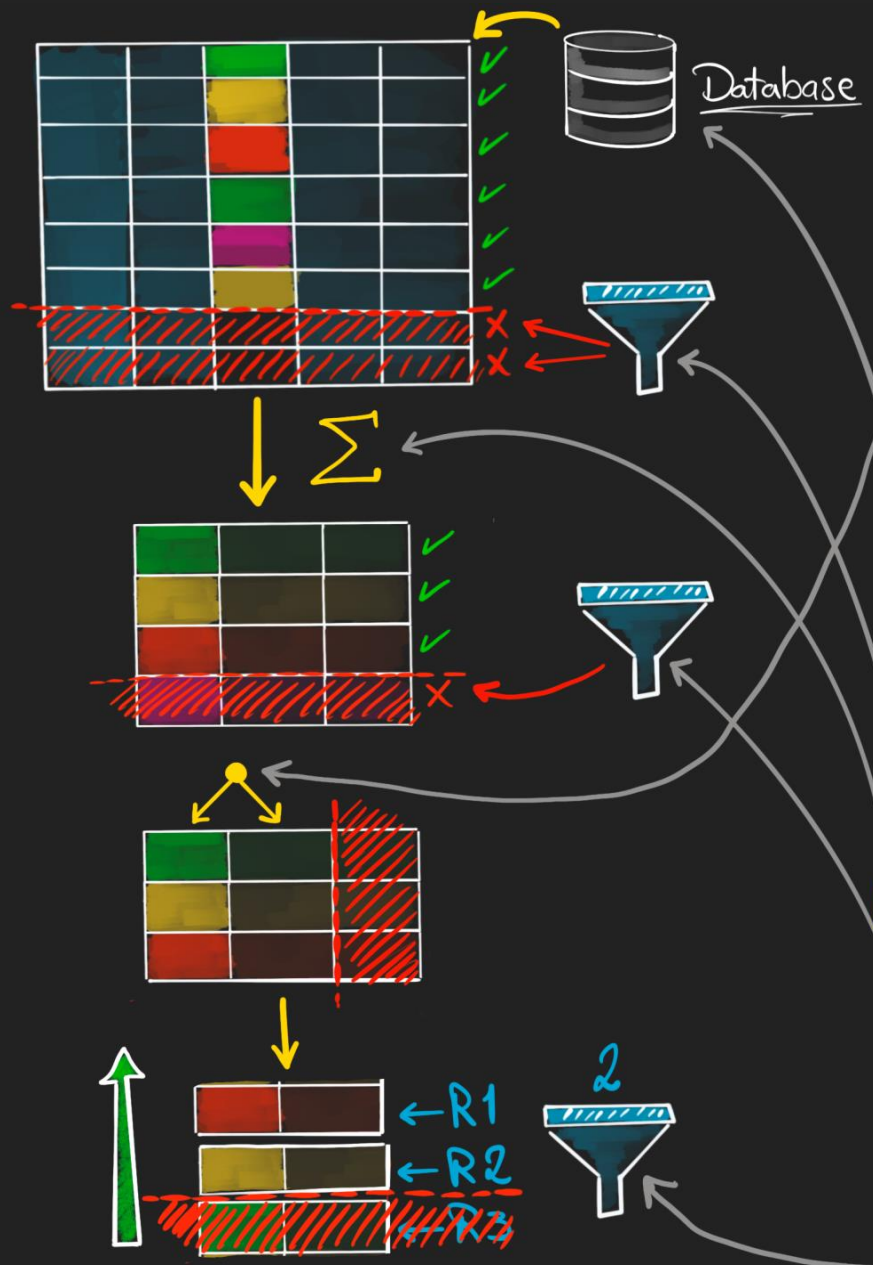
- Filter Columns
- Filter Duplicates
- Filter Result Rows
- Filter Rows Before Aggregations
- Filter Rows After Aggregations

Coding Order

```
SELECT DISTINCT TOP 2
    Col1,
    SUM(Col2)
FROM Table
WHERE Col = 10
GROUP BY Col1
HAVING SUM(Col2) > 30
ORDER BY Col1 ASC
```

Execute Order

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT DISTINCT
6. ORDER BY
7. TOP



Coding Order

5. SELECT DISTINCT Col1, SUM(Col2)
   TOP 2
   7
1. FROM Table
2. WHERE Col = 10
3. GROUP BY Col1
4. HAVING SUM(Col2) >30
6. ORDER BY Col1 ASC

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DATA WITH BARAA

BONUS

Sketches

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Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

① FROM

Result

② SELECT *

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	UK	750
4	Martin	Germany	500
5	Peter	USA	0

Execute Order

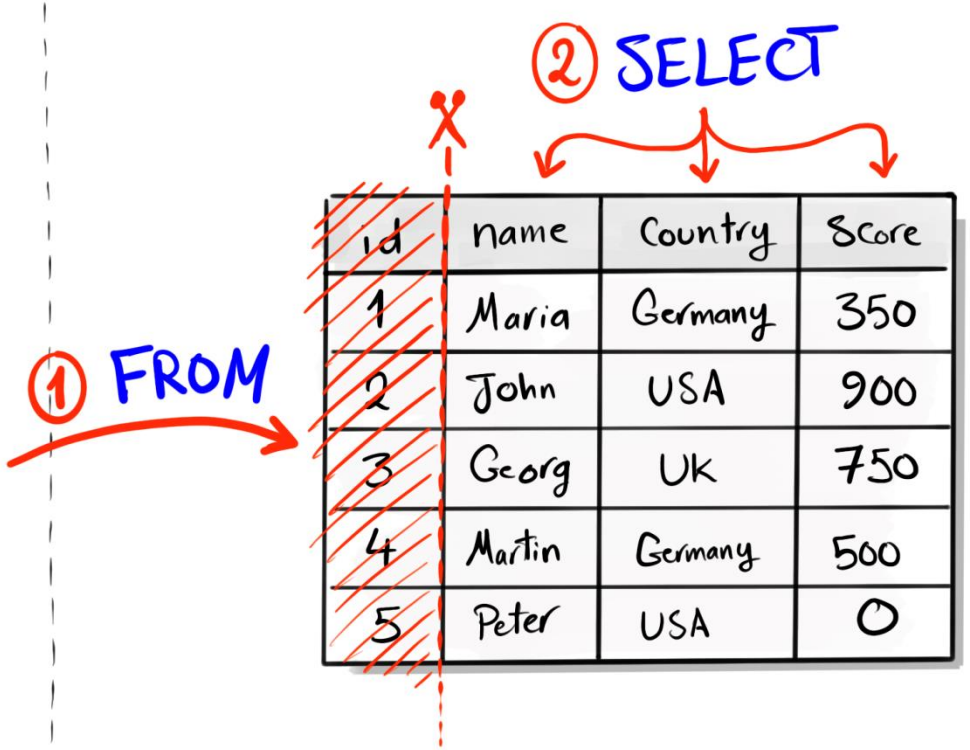
① FROM

DATA WITH BARAA

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



Result

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

Execute Order

1 FROM

2 SELECT

DATA WITH BARAA

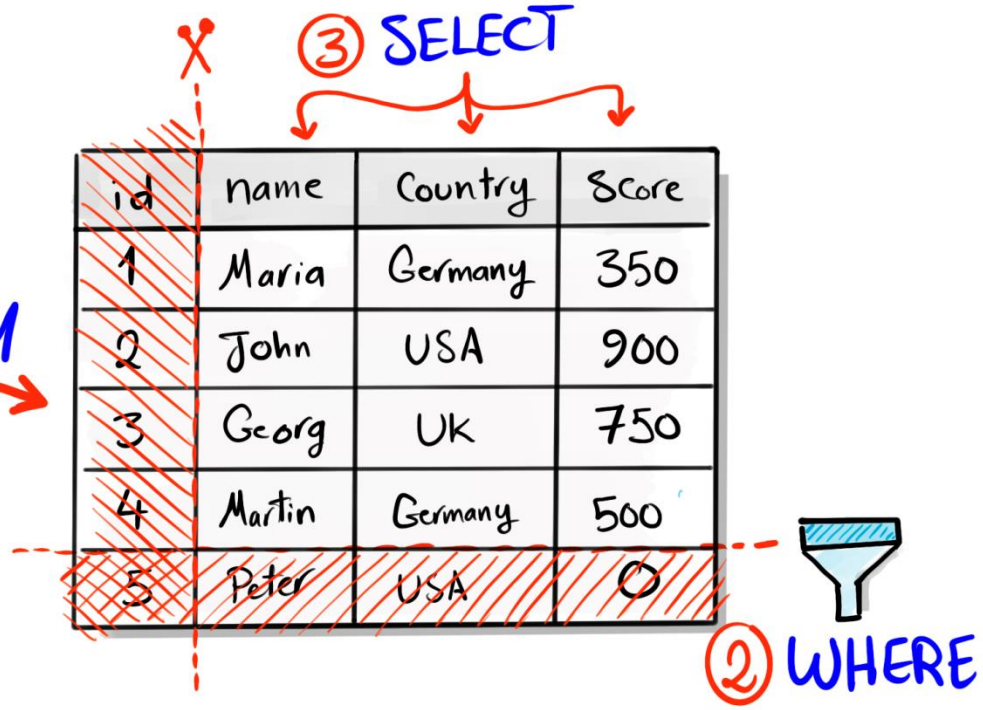
Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



Result



id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

Execute Order

1. FROM
2. WHERE
3. SELECT

DATA WITH BARAA

Name	score
Alex	90
Maria	50
Alex	30
Maria	80

DESC

Highest

Name	score
Alex	90
Maria	80
Maria	50
Alex	30

Lowest

ASC

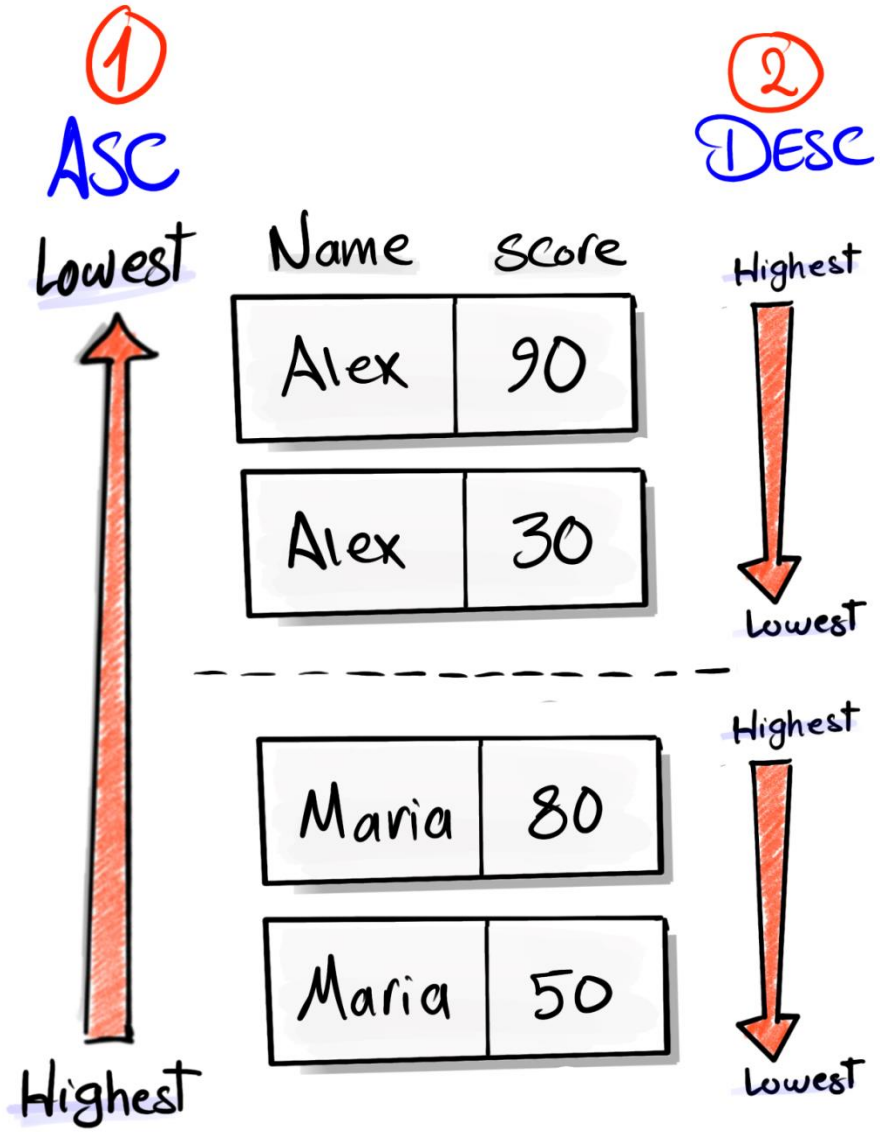
Lowest

Alex	30
Maria	50
Maria	80
Alex	90

Highest

DATA WITH BARAA

Name	score
Alex	90
Maria	50
Alex	30
Maria	80



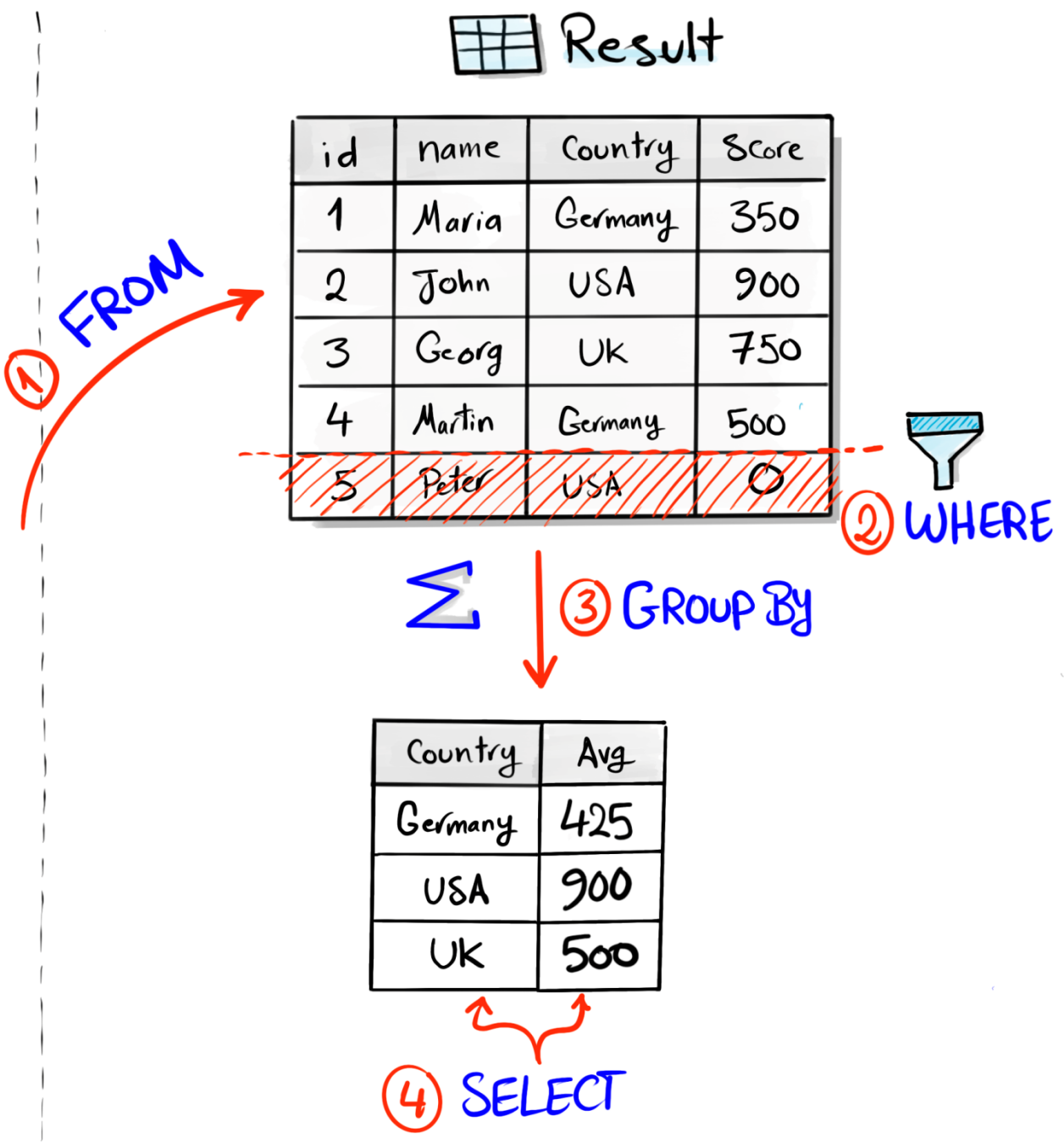
DATA WITH BARAA

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

Result



Execute Order

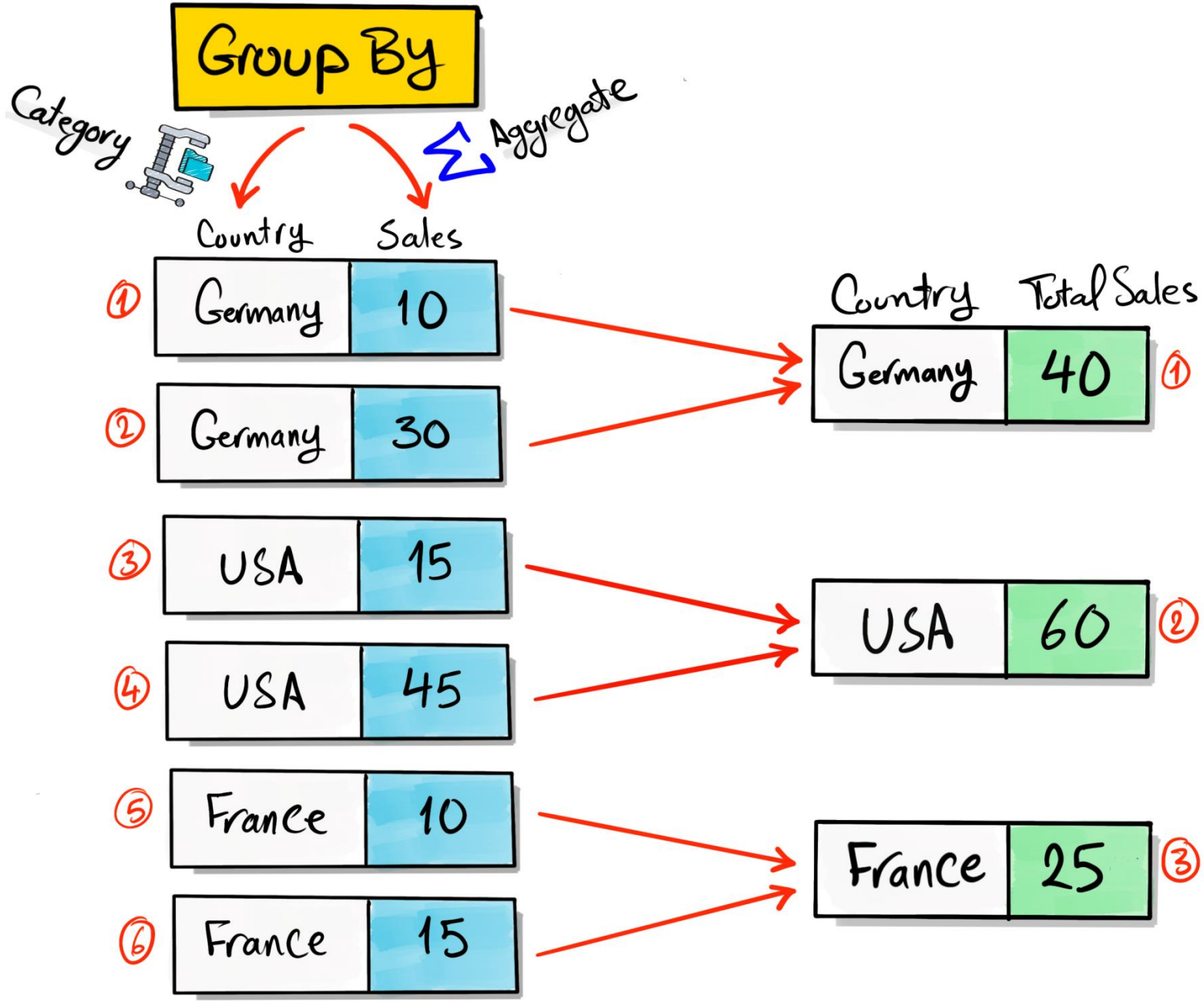
① FROM

② WHERE

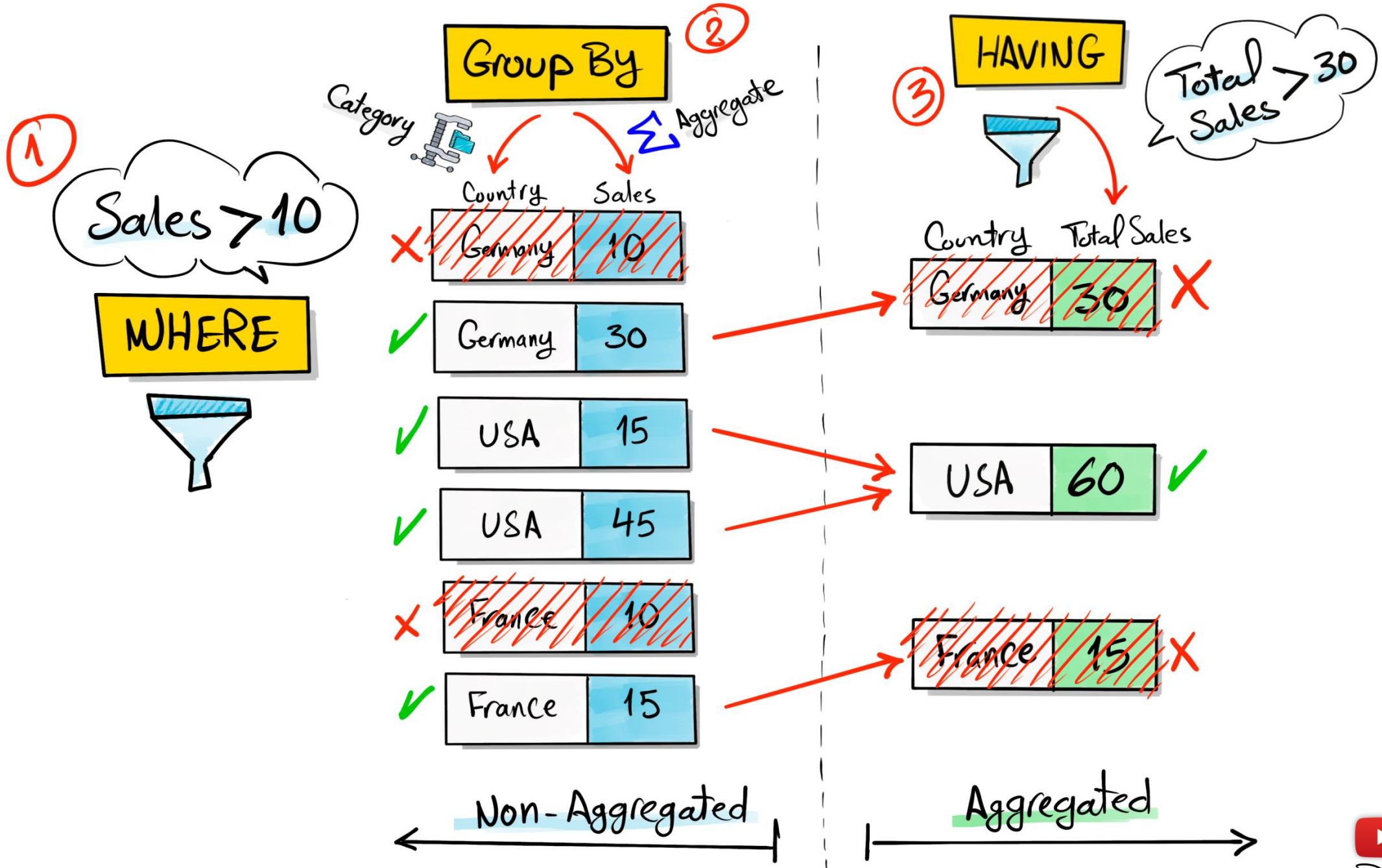
③ GROUP BY

④ SELECT

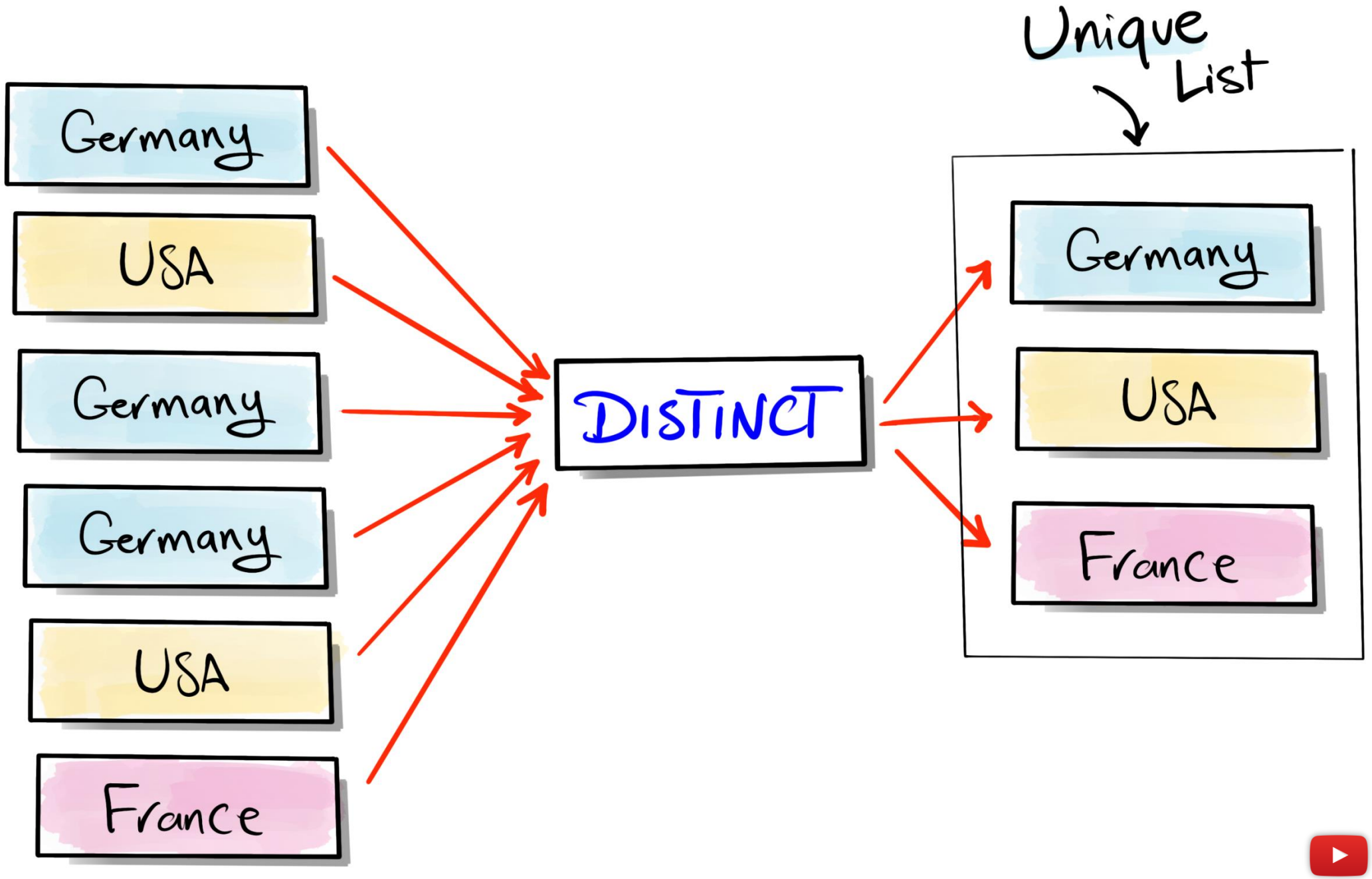
DATA WITH BARAA



DATA WITH BARAA



DATA WITH BARAA

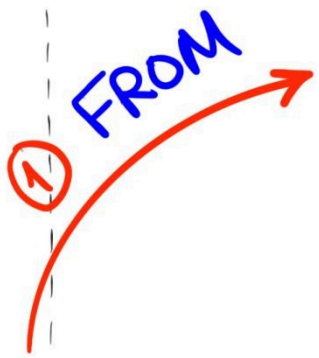


DATA WITH BARAA

Database

Customers

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0



Result

id	name	Country	Score
1	Maria	Germany	350
2	John	USA	900
3	Georg	Uk	750
4	Martin	Germany	500
5	Peter	USA	0

2 WHERE

Σ 3 GROUP BY

Country	Avg
Germany	425
USA	900
Uk	500

4 HAVING

5 SELECT

Country	Avg
UK	500
USA	900

6 ORDER BY ASC

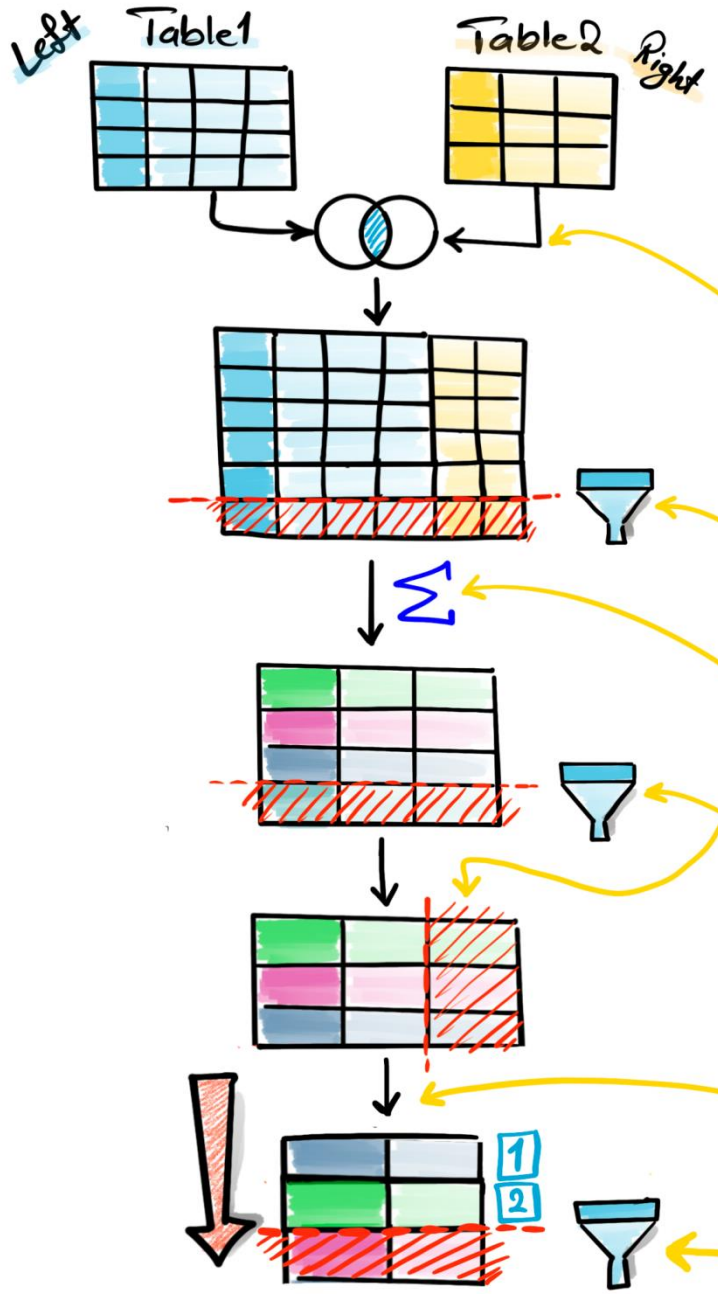
Execute Order

1. FROM
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT
6. ORDER BY

DATA WITH BARAA

Execution Order

1. FROM JOIN
2. WHERE
3. GROUP BY
4. HAVING
5. SELECT DISTINCT
6. ORDER BY
7. TOP



Coding Order

```
5  SELECT DISTINCT
       a.Col,
       SUM(b.Col)
   Top 2
7

1  FROM Table1 AS a
   JOIN Table2 AS b
     ON a.id = b.id

2  WHERE a.Col = 10

3  GROUP BY a.Col

4  HAVING SUM(b.Col)

6  ORDER BY a.Col
```


DATA WITH BARAA